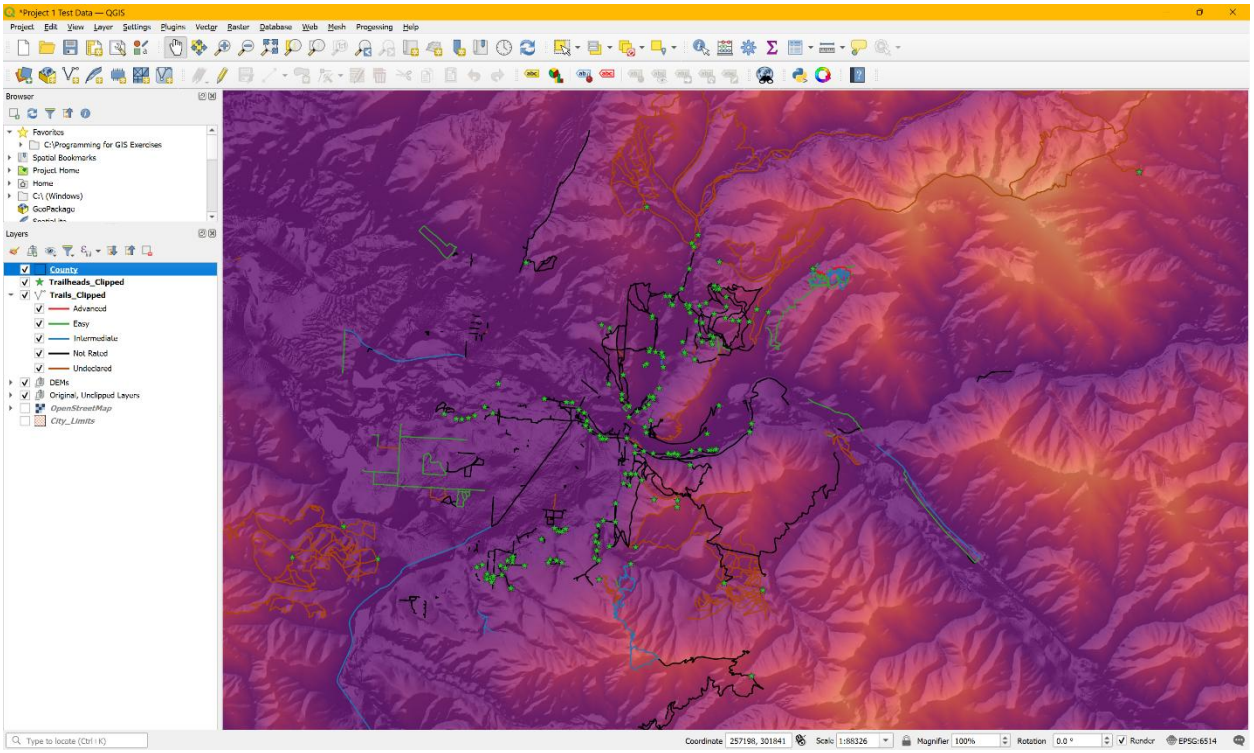
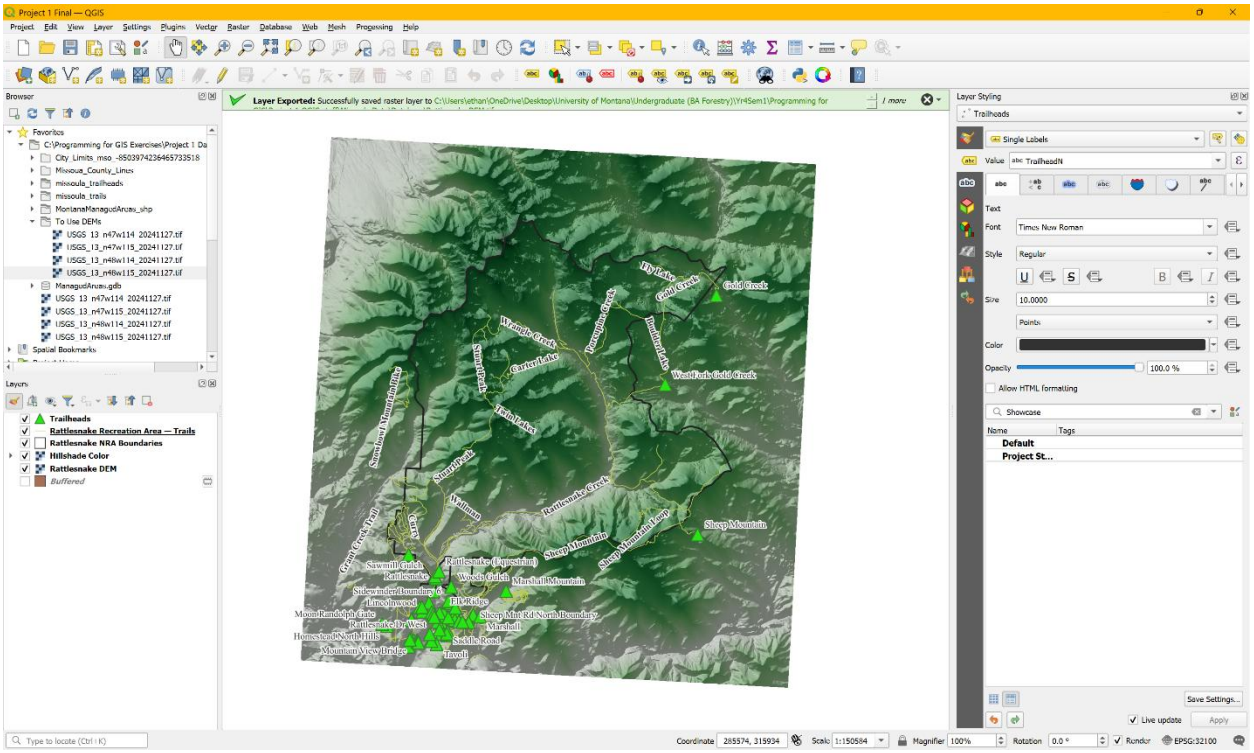
Project 1: Spatial Data Foundations

By Ethan Jones



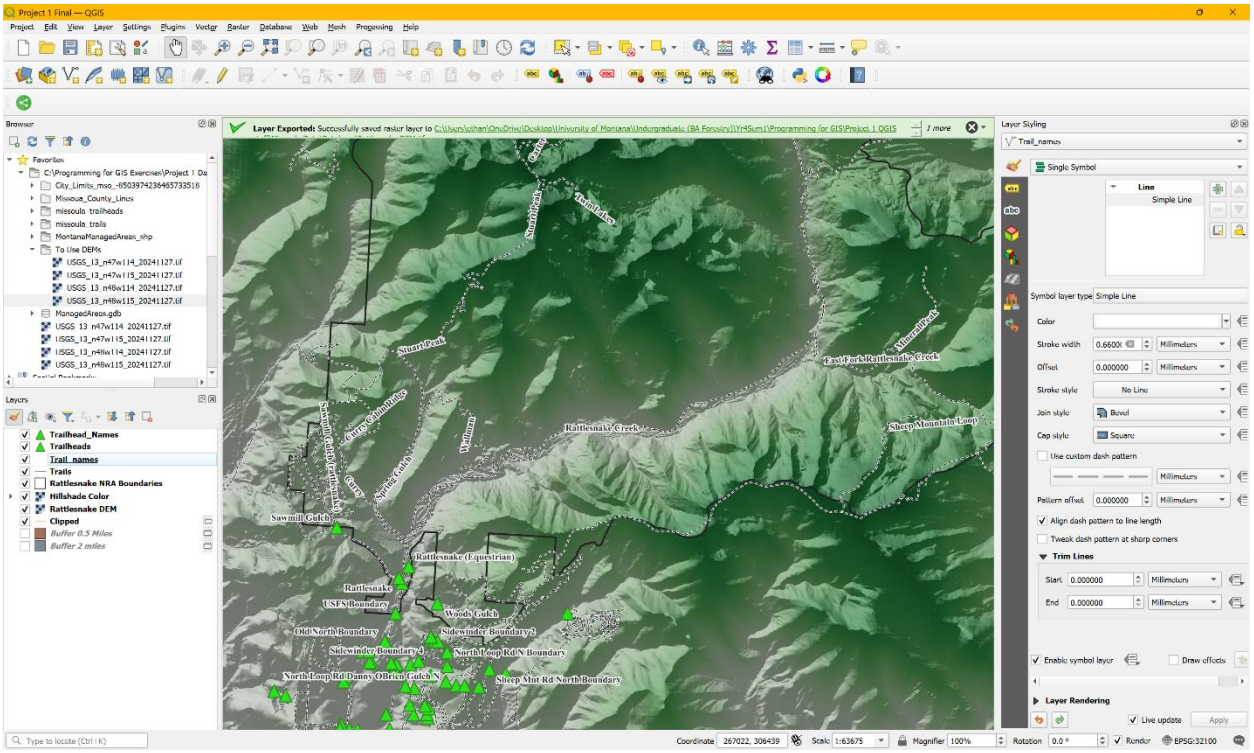
Development Photo 1:

This was the first iteration of the project, with the basic data being displayed uninformatively. Originally, my hypothesis was aimed at describing how population dispersion had led to an increase in recreational access to trailheads/trails. However, I had major difficulties pursuing this hypothesis and I wanted to focus on something more attainable, if not a little easier to accomplish both spatially and visually. The data in this photo is mostly untouched, and I was fiddling with DEM symbology.



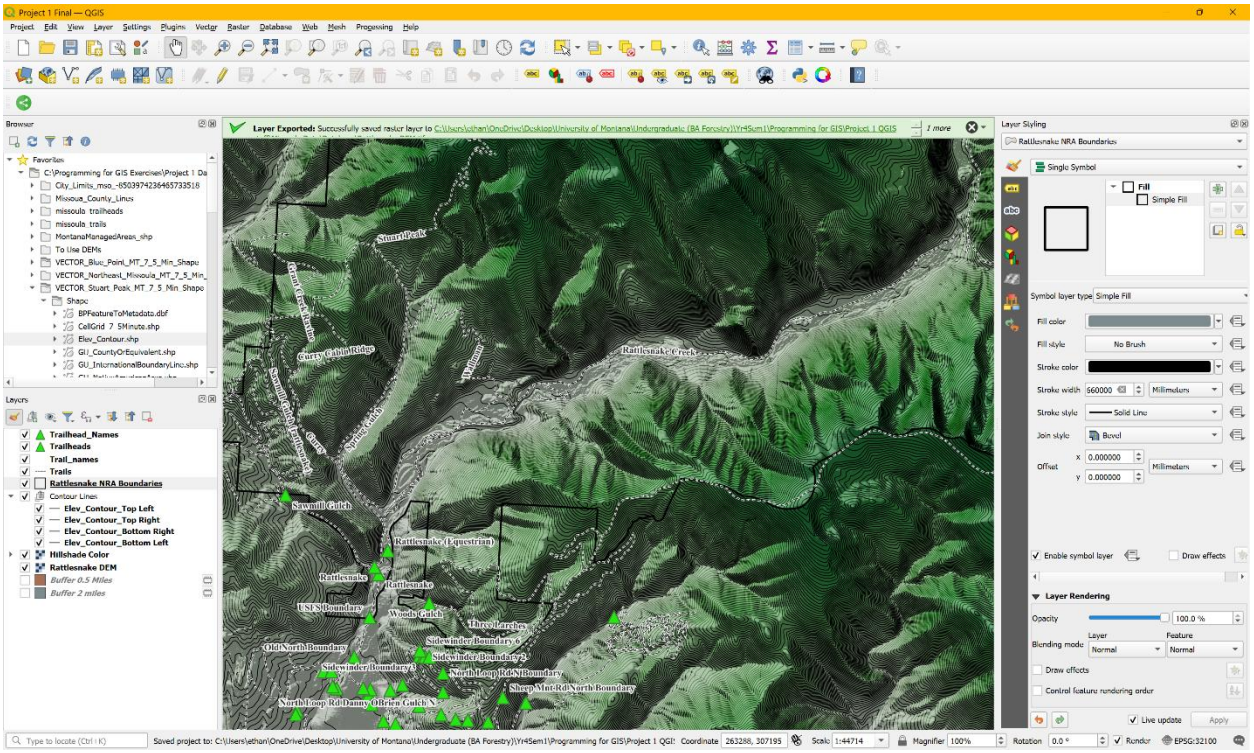
Development Photo 2:

In the second iteration of my project, I had revised my hypothesis and basically started again from scratch with the same data. This time my focus was on gauging trail difficulty within the rattlesnake recreation area based on elevation and distance data provided by my trail dataset. At this point I had buffered my vector and raster data to two miles outside of the rattlesnake recreation area's boundary and had begun to symbolize my DEM. My aim visually at this point was to make the map look similar to a typical topographic or tourist map by improving my trail visualization, DEM symbology, and adding contours.



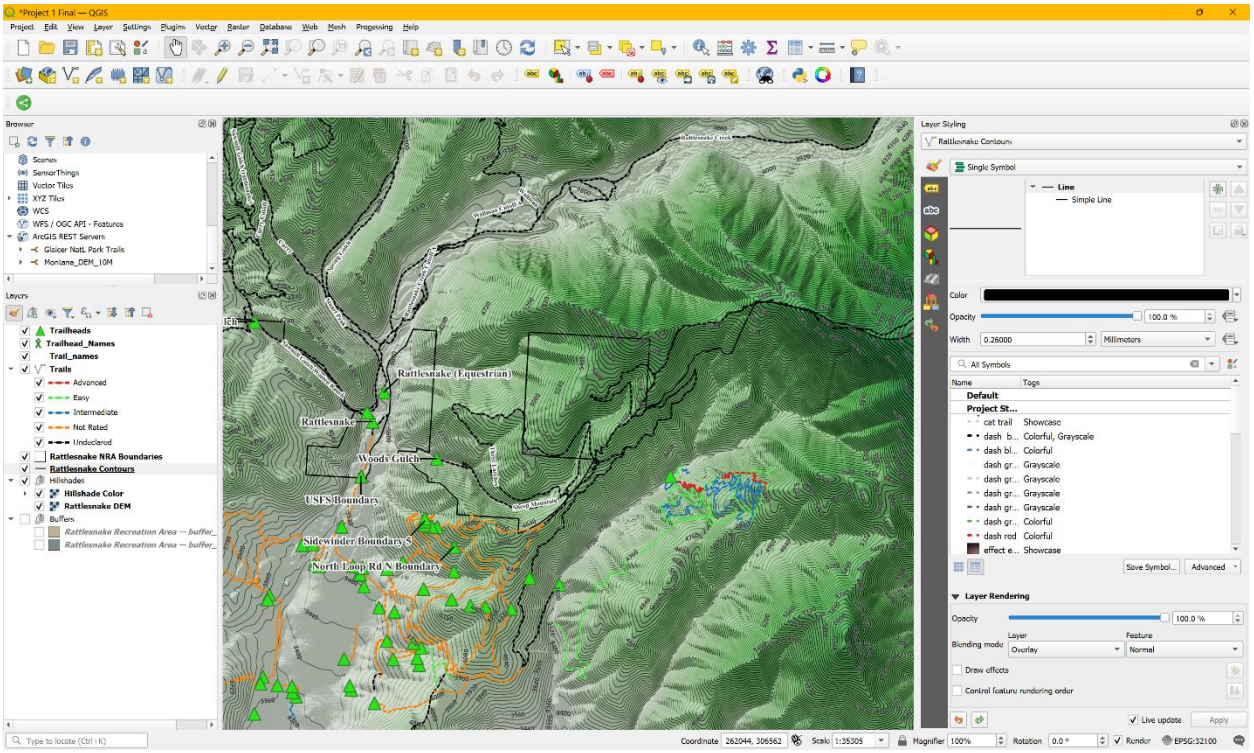
Development Photo 3:

In the third iteration of my project, I had narrowed down my spatial focus down to the lower rattlesnake and marshal mountain area. In this picture I had begun to start symbolizing the trails and trailheads to make them less intrusive on the overall map. My aim for development of my next iteration was to reduce cluttering of trailheads and trails on the southern part of my map by utilizing a buffer of half a mile to remove names and possibly some trails and trailheads. I also was actively searching for available contour data to add to my map.



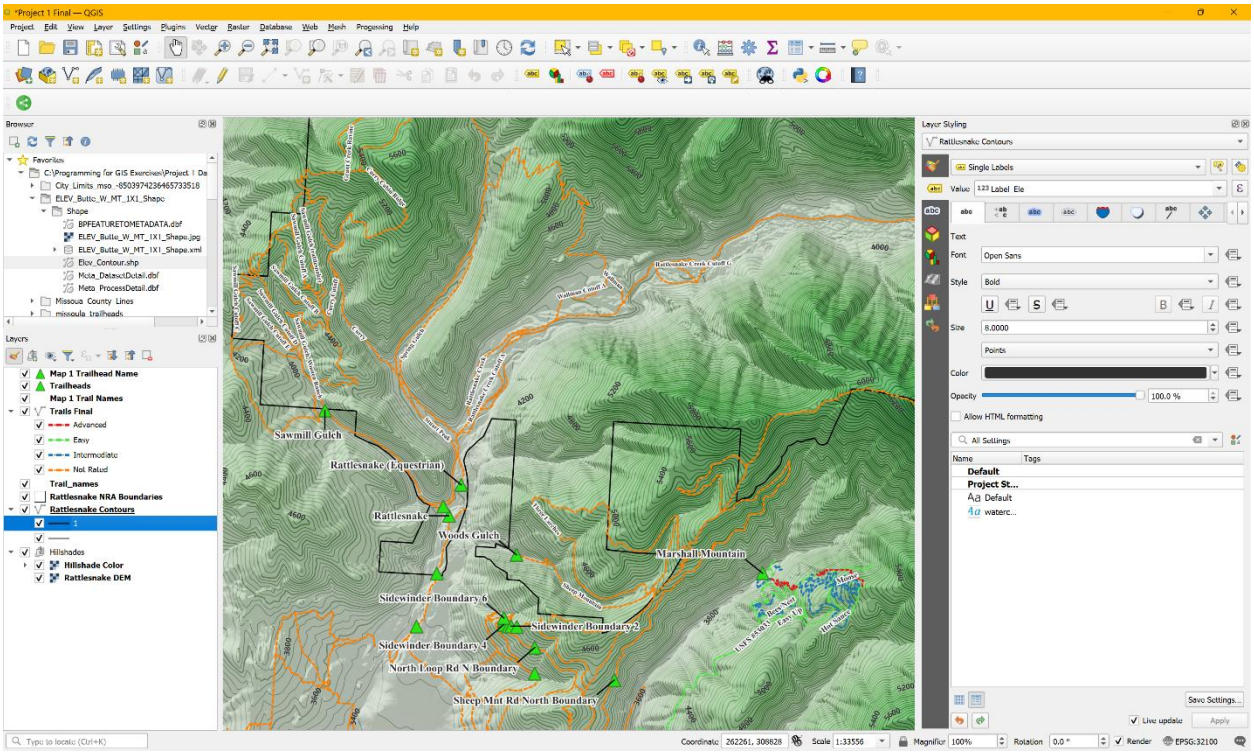
Development Photo 4:

In the fourth iteration of my project, I finally managed to find contour data on the USGS website that covered my spatial focus. I was somewhat disappointed by how it looked on the map, so I wanted to focus on making sure they blended into the map without being too visually demanding. At this point, I had settled on the symbology for my trails and DEM and just needed to classify my trails by difficulty.



Development Photo 5:

In the fifth iteration of my project, I had blended my contours using “Overlay”, categorized my trails by difficulty, and clipped more trail and trailhead names by the half a mile buffer. I had mostly left the difficulty grading of the trails untouched which is why there were the repeated classifications of unrated trails. By this point, we had our Data milestone, and I had received feedback from the class about how I could change my focus. Based on this, my focus shifted on trying to grade unrated trails based off existing trail difficulty ratings from the Marshall Mountain area. All that was left to do, visually, was to clean up the contours, naming, and to remove the repeated unrated class and merge them.



Development Photo 6:

In my final iteration of my project, I cleaned up the contours (by bolding and labeling every 200ft increase), removed insignificant trailheads, and named the Marshall Mountain area components. All that was left for me to do was to create my final maps and analyze the results.

Final Maps:

Hypothesis: can existing trail difficulty data be used in visual and analytical comparison to gauge the difficulty level of un-rated trails?

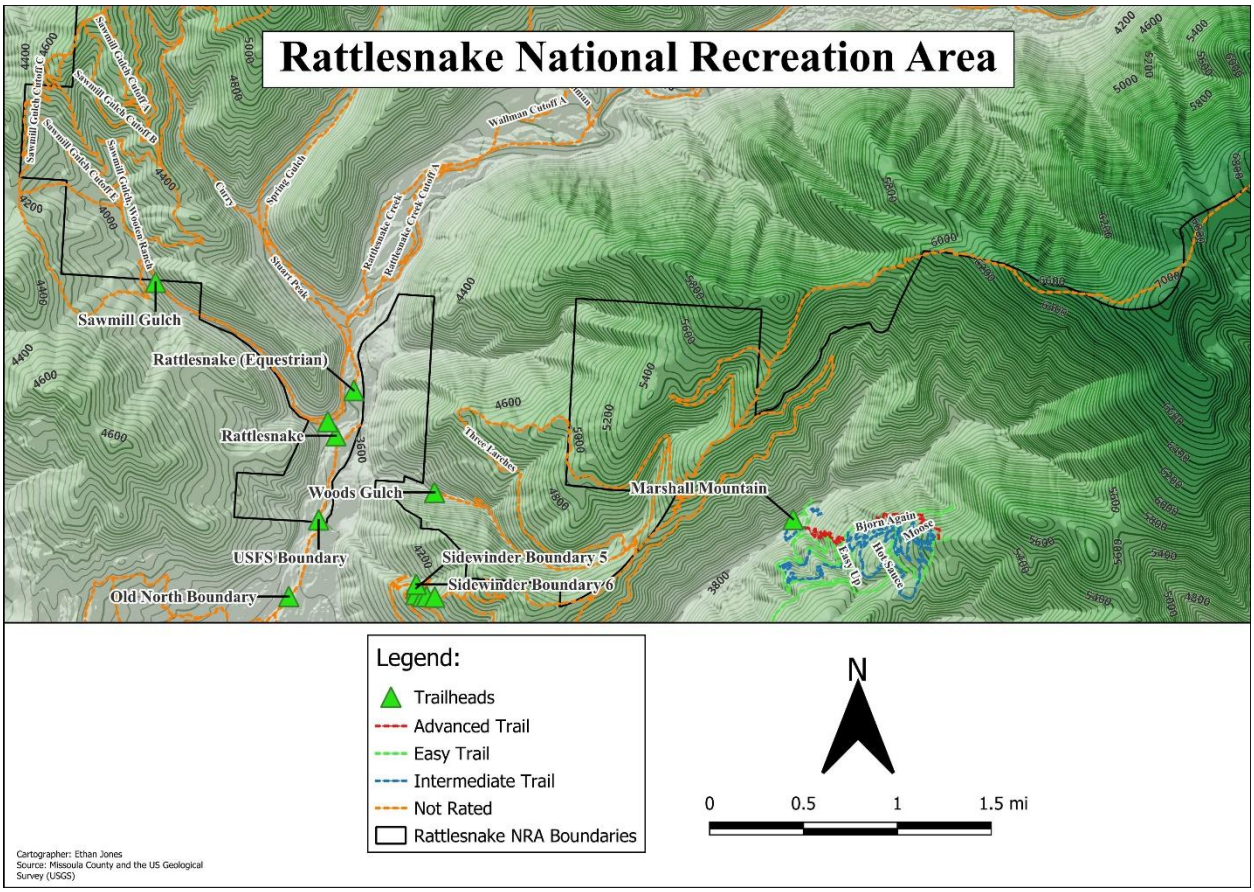


Figure 1: trail map of the Rattlesnake National Recreation Area and Marshall Mountain Trail systems. The focus areas are Marshall Mountain (bottom right) and Sawmill Gulch area of the Rattlesnake (top right)

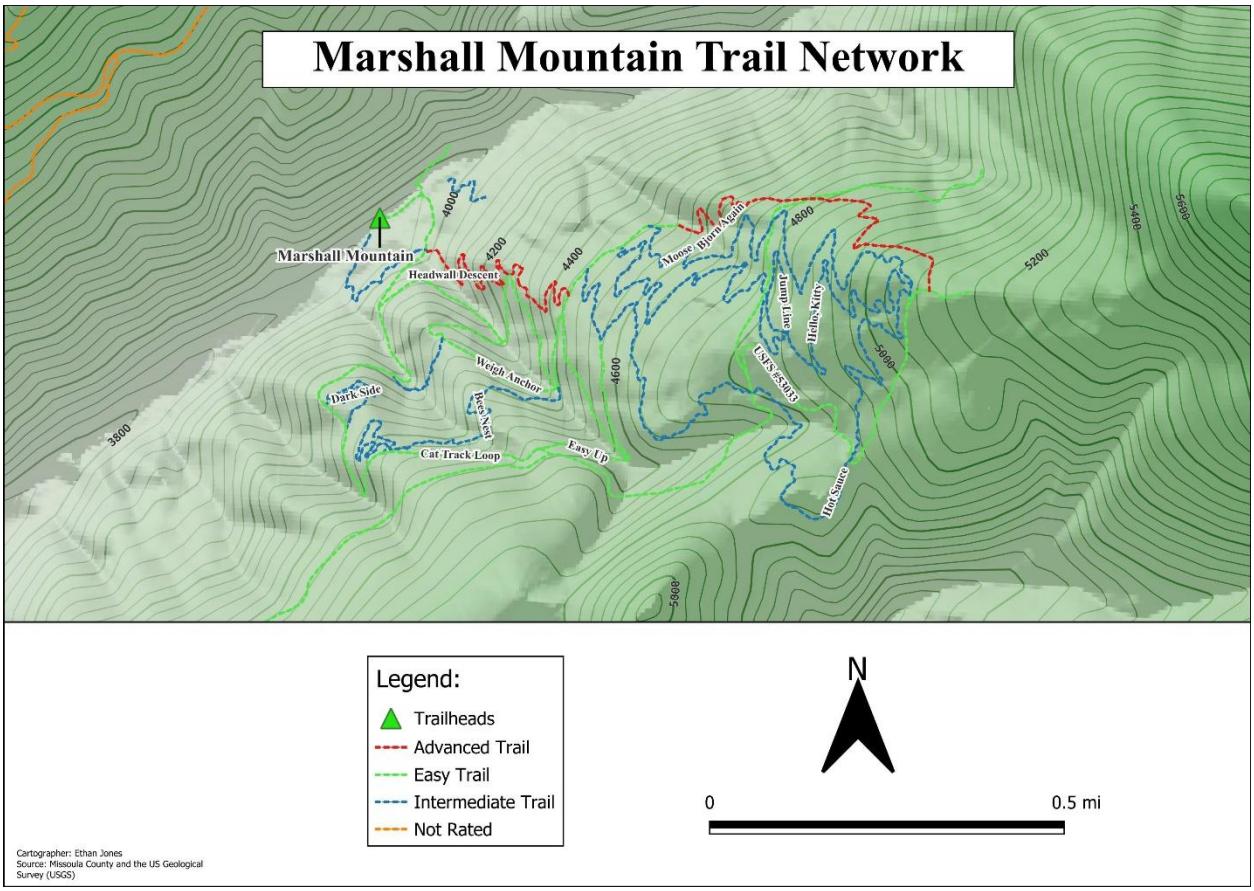


Figure 2: trail map of the Marshall Mountain trail network. The focus trails are Easy Up (bottom middle) and Hot Sauce (bottom right).

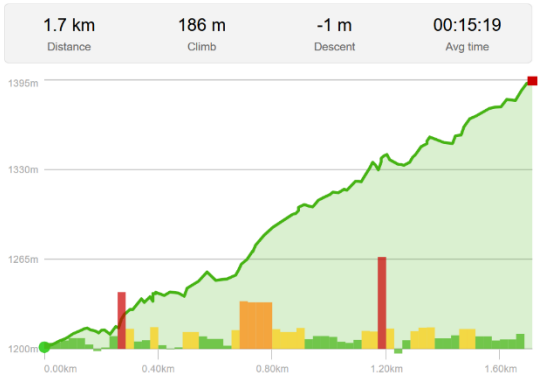


Figure 3. Easy Up elevation change map

Source: https://www.trailforks.com/trails/easy-up-458636/

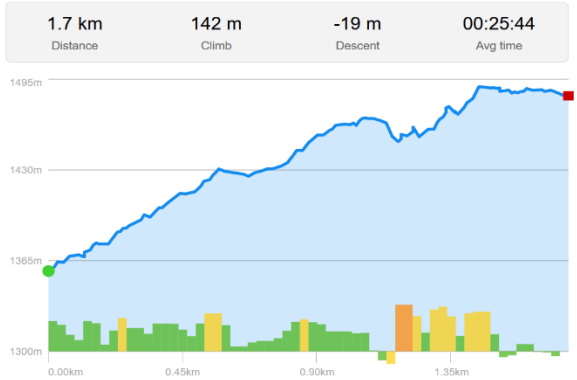


Figure 4. Hot Sauce elevation change map

Source: https://www.trailforks.com/trails/hot-sauce/

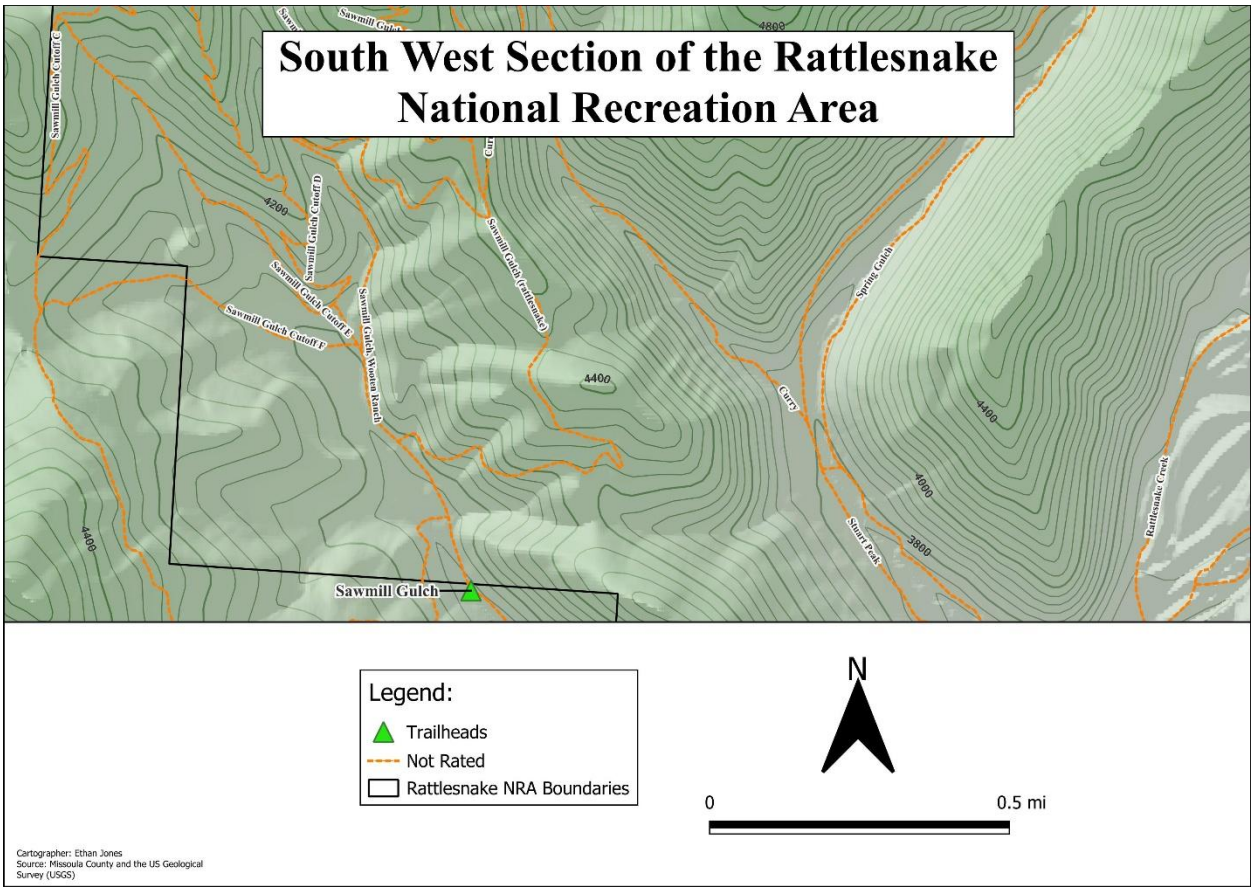


Figure 5: trail map of the southwest section of the Rattlesnake National Recreation Area trail network. The focus trails are Sawmill Gulch (rattlesnake) which is formally known as "The Burn" (middle) and Spring Gulch (top right).

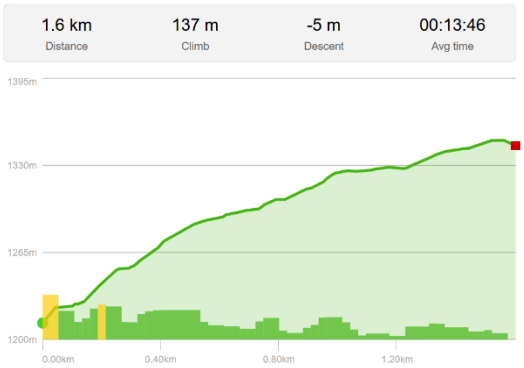


Figure 6. The Burn elevation change map

Source: https://www.trailforks.com/trails/the-burn-68659/

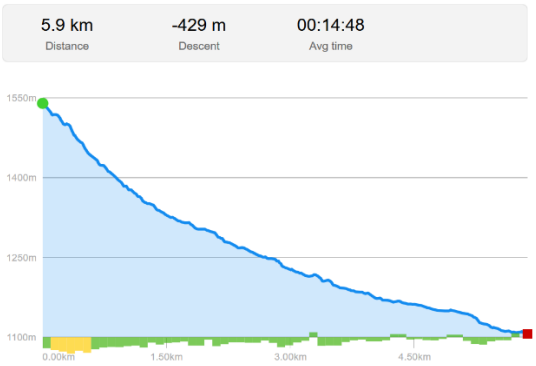


Figure 7. Spring Gulch elevation change map

Source: https://www.trailforks.com/trails/spring-gulch/

Conclusion:

Following the creation of my final set of maps, I was able to address my hypothesis in full. Visually, comparisons between the two sets of trails allowed me to inference some level of difficulty for the unrated trails. When comparing Easy Up to The Burn, you can see that both meander along the contour lines and have a large switchback component. I was able to infer that The Burn would probably be an easy or intermediate trail. However, comparing Hot Sauce to Spring Gulch proved more difficult. Hot Sauce had lots of twists and turns, and Spring Gulch was mostly straight. I would probably grade Spring Gulch an easy trail based off visuals. Comparing the elevation change graphs made assigning a difficulty value to the ungraded trails even more difficult, as the grading did not seem to carry over consistently, especially with the intermediate trails. Comparing the easy trails graphically seemed to match, as both roughly had the same elevation gain over roughly the same distance.

What I realized after comparing my maps, was that trail difficulty can be ambiguous and very subjective to area. This is especially the case when including trail complexity within a difficulty metric. For example, did my original trail difficulty dataset include technical skill required when determining the grade for the trail? Missoula County did not identify how they calculated the trail difficulty, which made comparison difficult. Although visual comparison between easy trails was relatively accurate, identifying intermediate difficulty and above proved challenging. Overall, I was able to address easy trails but not much above with my hypothesis. In future, I would like to have a better control group of data that clearly states how metrics like difficulty were determined, so that they could be repeated.

Visually, I am very pleased with how my maps turned out. I thoroughly enjoyed the process of exploring QGIS and the functionality that it offered.

Data Sources:

- Missoula County Trailheads and Trails: https://missoulamapscityofmissoula.hub.arcgis.com/search?categories=%252Fcategories%252Frecreation
- Rattlesnake National Recreation Area Boundaries: https://mslservices.mt.gov/Geographic_Information/Data/DataList/datalist_Details.aspx?did={80C2319F-17BC-4A67-B0DF-BB12B53D1D5E}
- USGS Data Downloads for DEMs and Contours: https://apps.nationalmap.gov/downloader/